

WASP-10b

R-0885

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_250805 · created 2026-07-29T01:08:54+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460892.8005 +/- 0.0068 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.086 +/- 0.061
Transit depth (Rp/Rs)^2	0.75 +/- 1.05 [%]
Orbital Inclination (inc)	84.53 +/- 1.9
Airmass coefficient 1 (a1)	0.71 +/- 0.38
Airmass coefficient 2 (a2)	0.29 +/- 0.43
Scatter in the residuals of the lightcurve fit is	53.61 %
Best Comparison Star	None
Optimal Aperture	0.75
Optimal Annulus	3.0
Transit Duration (day)	0.028 +/- 0.031

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.086 ± 0.061 vs 0.1592 (deviation 1.17σ)
Duration fit vs published	0.028 d vs 0.0946 d (deviation 2.1σ)
Mid-time O–C	12.0 min
Depth SNR	1.4
Residual scatter	53.61 %

Light curve

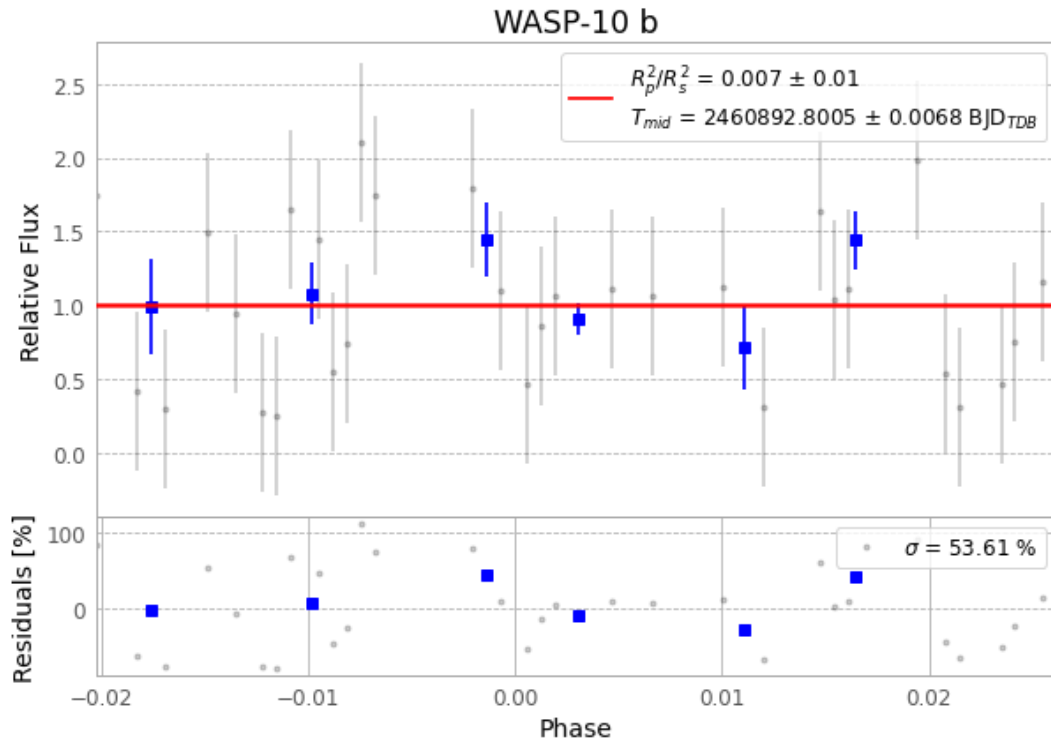


Figure 1 — FinalLightCurve_WASP-10 b_2025-08-04.png (SHA-256 e7f98ccdd9c4c994...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [189, 226], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 299811ee39a61b36...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

70 FITS frames (46 MB), WASP-10250805053315.FITS → WASP-10250805090615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-250805.log (54801a1cbc84...), FinalLightCurve_WASP-10 b_2025-08-04.png (e7f98ccdd9c4...), FinalLightCurve_WASP-10 b_2025-08-04.csv (a94e199bf4d6...)

Compute resources

Wall time 130.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 01:08Z.